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Article in *Computers & Chemical Engineering* · February 1991

DOI: 10.1016/0098-1354(91)87010-7

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## AN EFFICIENT IMPLEMENTATION OF NEWTON'S METHOD FOR COMPLEX NONIDEAL CHEMICAL EQUILIBRIA

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(Received 25 October 1989; final revision received 29 August 1990; received for publication 15 October 1990)

**Abstract**—An exact Newton method for the calculation of chemical equilibria involving nonideal solution phases is derived. It generalizes the classical Brinkley–Nasa–Rand approach for systems with ideal solution phases. Compared to existing algorithms it offers guaranteed local quadratic convergence to non-degenerate equilibria and computational efficiency for complex systems with many species.

### 1. INTRODUCTION

The modelling of complex multiphase chemical equilibria is fundamental to the understanding of many natural and technological processes [a general introduction is given in Smith and Missen (1982) and Gautam and Seider (1979)]. The simulation of such systems requires efficient and reliable algorithms for the computation of their equilibrium states, which occur as the minimum of the system's free energy. For complex systems, which may involve a large number of chemical species (mainly due to the gas phase) and a correspondingly large number of condensed phases (including solution phases), computational problems arise from the sheer number of variables, the necessity to determine the stable phase set out of a multitude of possible phase combinations and the nonideal nature of the solution phases (see e.g. Smith and Missen, 1982). In Greiner (1988b) we have presented an algorithm which allows to obtain a close estimate of the free energy minimum (including the stable phase set) effectively. Once this estimate is available, it can be refined by the application of Newton's method, which is known to converge quadratically to the exact solution. But for complex systems the application of Newton's method is hampered by the following facts:

1. The exact Newton correction has to be obtained from a system of  $(n + m)$  linear equations (the Newton equations), which arise from the linearization of the nonlinear equations defining the equilibrium (here  $n$  is the total number of component species and  $m$  is the number of chemical elements defining the system). Their solution may be computationally expensive or practically impossible for complex systems involving many species.

2. In the case that only ideal solution phases are involved, this computational problem can be avoided by eliminating the corrections of the species mole numbers from the Newton equations. The Newton correction can then be obtained from the solution of

$(m + p)$  linear equations for the increments of the Lagrange multipliers and the total mole numbers for each phase ( $p$  is the number of solution phases present at equilibrium). This method is the basis for the so called Brinkley–Nasa–Rand approach [see Smith and Missen (1982) Chap. 6] to the computation of equilibria with ideal phases. To extend this method to systems with nonideal phases one approximates the matrix of second derivatives of the free energies by their ideal parts and proceeds as in the Brinkley–Nasa–Rand algorithm (for the first derivatives the nonideal values are used). This “intermediate” approach [cf. Smith and Missen, (1982) Chap. 7] has been adopted in a number of equilibrium codes (Eriksson and Rosen, 1973; Eriksson and Hack, 1990; Gautam and Seider, 1979). But already simple examples from thermodynamics demonstrate that this method suffers from severe convergence problems for systems with nonideal phases even if a very close estimate of the true solution is available [cf. for instance Michelsen (1982) p. 29]. The problem of (practical) nonconvergence of the ideal approximation method is akin to the well-known convergence problems for steepest descent methods (Gill and Wright, 1981).

In this paper we show that it is possible to implement the Brinkley–Nasa–Rand algorithm for non-ideal systems as an exact Newton method in which the second derivatives of the free energies are taken into account without approximation. Specifically we prove that the exact Newton equations can be reduced to  $(m + p)$  linear equations provided we are dealing with a nondegenerate equilibrium (for degenerate equilibria, special measures which are described in Section 5 have to be taken). The argument is based on the observation that the matrix of second derivatives of the modified free energies:

$$\bar{G}(n_1, \dots, n_s) = G(n_1, \dots, n_s) + \left( \sum_{i=1}^s n_i \right) \log \left( \sum_{i=1}^s n_i \right),$$

of a stable phase with component mole numbers  $(n_1, \dots, n_s)$  and free energy  $G(n_1, \dots, n_s)$  must be positive-definite for reasons of thermodynamic stability.

The advantages of the scheme are obvious: as we apply Newton's method exactly we have guaranteed quadratic local convergence. Yet the computational effort remains manageable even for the most complex systems.

The presentation is organized as follows: in Section 2 we give the mathematical formulation of the problem. Section 3 discusses the application of Newton's method to the solution of the nonlinear equilibrium equations. A modified form of the equilibrium equations is introduced and it is demonstrated that the increments of the mole numbers of each solution phase can be eliminated from the corresponding Newton equations. Various strategies for the solution of the linearized equations are discussed.

Questions related to the numerical implementation of our method are considered in Section 4. Finally there is a conclusion and Appendices giving the mathematical proofs and an example highlighting the weaknesses of the ideal approximation method.

## 2. MATHEMATICAL FORMULATION

Chemical equilibria are defined by the minimum of the chemical system's free energy, which depends on the mole numbers of each individual chemical species in the system.

In a system with  $p$  different phases ( $j = 1, \dots, p$ ) the chemical equilibrium problem can be formulated as follows:

$$G(\mathbf{n}^1, \dots, \mathbf{n}^p) = \sum_{j=1}^p G^j(\mathbf{n}^j) = \min, \quad (1a)$$

subject to the restrictions (mass balances):

$$\mathbf{R}\mathbf{n} = \sum_{j=1}^p \mathbf{R}^j \mathbf{n}^j = \mathbf{b} \quad \mathbf{n}^j \geq 0, \quad j = 1, \dots, p. \quad (1b)$$

Here the following notation is assumed:

$$\mathbf{n}^j = (n_1^j, \dots, n_{s(j)}^j),$$

are the mole numbers of the  $s(j)$  species constituting the  $j$ th phase with free energy  $G^j$  and  $\mathbf{n} = (\mathbf{n}^1, \dots, \mathbf{n}^p)$ . The constraint matrix  $\mathbf{R}$  is given by  $(\mathbf{R}^1, \dots, \mathbf{R}^p)$ , where  $\mathbf{R}^j = (\mathbf{R}_1^j, \dots, \mathbf{R}_{s(j)}^j)$  and the  $\mathbf{R}_i^j$  are column vectors from  $\mathbf{R}^m$ .  $\mathbf{b}$  is a vector from  $\mathbf{R}^m$ .

For pure condensed phases, the free energy:

$$G(n) = \mu^0 n,$$

where  $\mu^0$  is the chemical potential, is a linear function of the mole number  $n$ .

As our concern is local convergence to stable equilibria, we may assume that all solution phases are stable, i.e. that their free energy is a convex function of the mole numbers of the constituting species [ $s(j) > 1$ ]. It is expressed as:

$$G(n_1, \dots, n_s) = \sum_{i=1}^s n_i \{ \mu_i^0 + \ln[a_i(\mathbf{x})] \}.$$

Here  $\mathbf{x}$  designates the composition of the phase:

$$\mathbf{x} = (x_1, \dots, x_s) = (n_1/N, \dots, n_s/N)$$

where

$$N = \sum_{i=1}^s n_i,$$

is the total mole number of the phase. The functions  $a_i(\mathbf{x})$  are the activities of the component species of the solution phase and the  $\mu_i^0$  are their chemical potentials. The energy is expressed in units of  $RT$ .

Solution phases with activities of the form  $a_i(\mathbf{x}) = x_i$ ,  $i = 1, \dots, s$  are called ideal.

The excess part of the free energy is defined by:

$$G_{\text{ex}}(n_1, \dots, n_s) = G(n_1, \dots, n_s) - \sum_{i=1}^s n_i [\mu_i^0 + \ln(x_i)].$$

For more information on these matters see Smith and Missen (1982) and Greiner (1988a).

To compute a minimum for problem (1) it is advisable to proceed in two stages:

1. In the first stage an estimate of the minimum is calculated. In particular, the stable phases present at equilibrium are determined out of a multitude of possible phase combinations.
2. With the estimate as a starting point a Newton method is used to obtain the accurate solution.

As we have shown in Greiner (1988b) the generalized linear programming algorithm is very efficient in furnishing a close estimate of the solution. Therefore we assume in the sequel that the phases present at equilibrium are known. Furthermore we do assume that all phases are solution phases as the mole numbers of pure condensed phases can always be eliminated from problem (1) via the mass balance equations (1b). This follows readily from the fact that one can always find a representation of the constraint matrix  $\mathbf{R}$  in which the column vectors corresponding to the pure condensed phases are linearly independent unit vectors, see Schnedler (1984). By eliminating the pure species from the problem one avoids the numerical singularities associated with their occurrence [Smith and Missen (1982) Chap. 9.2.1].

The following general assumptions concerning problem (1) are implied by general thermodynamics (Smith and Missen, 1982; Greiner, 1988a):

1. The matrix  $(\mathbf{R}^1, \dots, \mathbf{R}^p)$  has maximal rank.
2. The functions  $G^j$ ,  $j = 1, \dots, p$  are homogeneous and can be written as:

$$G^j(n_1, \dots, n_s) = \left( \sum_{i=1}^{s(j)} n_i \right) \bar{G}^j(x_1, \dots, x_s).$$

Here  $\bar{G}^j$  is the molar free energy of the  $j$ th phase ( $j = 1, \dots, p$ ).

3. The functions:

$$\bar{G}^j : \Delta^{s(j)} = \left\{ \mathbf{x} \mid \sum_{i=1}^{s(j)} x_i = 1, x_i \geq 0 \right\} \rightarrow R$$

are strictly convex (i.e. the phases  $G^j$  are stable) or equivalently the matrices  $\nabla^2 G^j$  are positive-definite when restricted to the subspaces:

$$\mathbf{V}_0^{s(j)} = \left\{ \mathbf{v} \in \mathbf{R}^{s(j)} \mid \sum_{i=1}^{s(j)} v_i = 0 \right\}.$$

As a consequence of these assumptions the following statements obtain:

*Lemma 1*

- (a)  $\nabla^2 G^j(\mathbf{n}^j) \mathbf{n}^j = \mathbf{0}$  for  $j = 1, \dots, p$ ;
- (b) the matrices  $\nabla^2 G^j(\mathbf{n}^j)$  are positive semidefinite for  $j = 1, \dots, p$ .

A formal proof of Lemma 1 is provided in Lemma A2 in Appendix A. We remark that (a) is a direct consequence of the homogeneity of the  $G^j$  and is known as the Gibbs–Duhem relation in the chemical literature.

The optimality conditions for a minimum of problem (1) are:

$$\nabla G(\mathbf{n}) + \tilde{\mathbf{R}}\mathbf{w} = \mathbf{0} \quad \text{or} \quad \nabla G^j(\mathbf{n}^j) + \tilde{\mathbf{R}}^j\mathbf{w} = \mathbf{0} \quad \text{for } j = 1, \dots, p \quad (2a)$$

and

$$\mathbf{R}\mathbf{n} - \mathbf{b} = \mathbf{0} \quad \text{or} \quad \sum_{j=1}^p \mathbf{R}^j \mathbf{n}^j - \mathbf{b} = \mathbf{0}. \quad (2b)$$

Here  $\tilde{\mathbf{R}}$  denotes the transpose of the matrix  $\mathbf{R}$ .

In this investigation we want to analyze Newton's method for the solution of nonlinear equations (2). As explained above we may suppose that a good estimate of the solution is already available. In particular we assume that the phases occurring at equilibrium are known and that the pure phases have been eliminated from the problem. Hence all the phases  $G^j$  ( $j = 1, \dots, p$ ) in problem (1) are solution phases.

Let  $\mathbf{n} = (\mathbf{n}^1, \dots, \mathbf{n}^p)$  and  $\mathbf{w} = (w_1, \dots, w_m)$  denote an estimate for the solution of equations (2). To improve this estimate one linearizes equations (2) with respect to  $\mathbf{n}$  and  $\mathbf{w}$

$$\nabla^2 G(\mathbf{n}) \Delta \mathbf{n} + \tilde{\mathbf{R}} \Delta \mathbf{w} = -\nabla G(\mathbf{n}) - \tilde{\mathbf{R}}\mathbf{w} \quad (3a)$$

or

$$\nabla^2 G^j(\mathbf{n}^j) \Delta \mathbf{n}^j + \tilde{\mathbf{R}}^j \Delta \mathbf{w} = -\nabla G^j(\mathbf{n}^j) - \tilde{\mathbf{R}}^j \mathbf{w} \quad \text{for } j = 1, \dots, p$$

and

$$\mathbf{R} \Delta \mathbf{n} - \mathbf{b} - \mathbf{R}\mathbf{n} \quad \text{or} \quad \sum_{j=1}^p \mathbf{R}^j \Delta \mathbf{n}^j = \mathbf{b} - \sum_{j=1}^p \mathbf{R}^j \mathbf{n}^j. \quad (3b)$$

In order for Newton's method to be applicable the linear equations must have a unique solution. The following proposition states that this is always true for "nondegenerate" equilibria, for which the total mole numbers and composition of the phases are uniquely determined.

*Proposition 1*

For  $G^j$   $j = 1, \dots, p$  strictly convex, the following statements are equivalent:

1. The linear equations (3) have a unique solution  $\Delta \mathbf{n}$  and  $\Delta \mathbf{w}$ .
2. The vectors  $\mathbf{R}^j \mathbf{n}^j$  ( $j = 1, \dots, p$ ) are linearly independent.
3. The equilibrium is nondegenerate, i.e.

$$\Delta \tilde{\mathbf{n}} \nabla^2 G(\mathbf{n}) \Delta \mathbf{n} > 0 \quad \text{for all } \Delta \mathbf{n} \neq \mathbf{0} \text{ with } \mathbf{R} \Delta \mathbf{n} = \mathbf{0}.$$

For proof see Proposition A1 in Appendix A. Proposition 1 establishes that a minimum is nondegenerate if and only if Newton's equations for this minimum are nonsingular.

If the initial iterate  $\mathbf{n}$  for the solution of equations (3) is feasible (i.e. if it satisfies the constraints  $\mathbf{R}\mathbf{n} = \mathbf{b}$ ), the same is true for all subsequent iterates. In particular, if we start with a feasible iterate  $\mathbf{n}$  the correction  $\Delta \mathbf{n}$  may be viewed as the solution of the quadratic minimization problem:

$$\frac{1}{2} \Delta \tilde{\mathbf{n}} \nabla^2 G(\mathbf{n}) \Delta \mathbf{n} + \nabla G(\mathbf{n}) \Delta \mathbf{n} + G(\mathbf{n}) = \min,$$

subject to

$$\mathbf{R} \Delta \mathbf{n} = \mathbf{0}.$$

Consequently  $\Delta \mathbf{n}$  may be considered as a descent direction for problem (1) based on a quadratic approximation. Therefore Newton's method for the solution of equations (2) can be regarded as a minimization method with fixed stepsize 1. But as we are only interested in local convergence, the distinction is irrelevant (for Newton's minimization method the stepsize converges to 1).

### 3. SOLUTION STRATEGIES FOR NEWTON'S EQUATIONS

To apply Newton's method to the calculation of nondegenerate equilibria one has to solve the system of linear equations:

$$\begin{pmatrix} \nabla^2 G & \tilde{\mathbf{R}} \\ \mathbf{R} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \Delta \mathbf{n} \\ \Delta \mathbf{w} \end{pmatrix} = \begin{pmatrix} -\nabla G - \tilde{\mathbf{R}}\mathbf{w} \\ \mathbf{b} - \mathbf{R}\mathbf{n} \end{pmatrix}, \quad (4)$$

with  $(n+m)$  unknowns. But for large systems the direct solution of these equations is completely impractical and methods which take advantage of the special structure of the coefficient matrix (4) are called for. Basically there are two possible approaches [see Gill and Wright (1981), Heath (1978) and Nash (1985)].

*Null space methods*

Represent the affine subspace:

$$\{\Delta \mathbf{n} \mid \mathbf{R} \Delta \mathbf{n} = \mathbf{b} - \mathbf{R}\mathbf{n}\}$$

as

$$\{\Delta \mathbf{n} = \mathbf{n}^* + \mathbf{Z}\mathbf{d} \mid \mathbf{d} \in \mathbf{R}^{n-m}\},$$

where  $\mathbf{Rn} = \mathbf{b} - \mathbf{Rn}$  and  $\mathbf{Z}$  is a  $(n, n-m)$  matrix. Multiplying equation (3a) from the left with  $\mathbf{Z}$  and introducing  $\Delta \mathbf{n} = \mathbf{n}' + \mathbf{Zd}$  leads to:

$$\mathbf{Z}(\nabla^2 G)\mathbf{Zd} = \mathbf{Z}[-\nabla G - (\nabla^2 G)\mathbf{n}'], \quad (5a)$$

(NB.  $\mathbf{Z}\mathbf{R} = \mathbf{0}$ ). As equations (4) are assumed to be nondegenerate, the matrix  $\mathbf{Z}(\nabla^2 G)\mathbf{Z}$  is positive-definite (cf. the proof of Proposition 1 in Appendix A). Hence (5a) can be solved for  $\mathbf{d}$ , which gives  $\Delta \mathbf{n} = \mathbf{n}' + \mathbf{Zd}$  and

$$\mathbf{R}\Delta \mathbf{w} = -(\nabla^2 G)\mathbf{Zd} - \nabla G - \mathbf{R}\mathbf{w} - (\nabla^2 G)\mathbf{n}', \quad (5b)$$

(NB. as  $\mathbf{Z}$  annuls the right-hand side of this equation,  $\Delta \mathbf{w}$  is well-defined.)

For large systems null space methods are not practical as they require the solution of  $n-m$  linear equations with a dense matrix. Such systems can only be solved iteratively (Gill and Wright, 1981; Nash, 1985). In chemical engineering literature null-space methods are termed stoichiometric (Smith and Missen, 1982). These authors also discuss methods for ideal systems, which are based on sparse approximations to equations (5).

#### Range space methods

These methods require that  $\nabla^2 G$  is nonsingular, i.e. in our instance positive-definite. In this case one can rewrite equations (3a) as:

$$\Delta \mathbf{n} = (\nabla^2 G)^{-1}[-\nabla G(\mathbf{n}) - \mathbf{R}\mathbf{w} - \mathbf{R}\Delta \mathbf{w}]. \quad (6a)$$

Introducing this expression into (3b) gives  $m$  equations for  $\Delta \mathbf{w}$ :

$$\begin{aligned} [\mathbf{R}(\nabla^2 G)^{-1}\mathbf{R}]\Delta \mathbf{w} \\ = -\mathbf{b} + \mathbf{Rn} - \mathbf{R}(\nabla^2 G)^{-1}[\nabla G(\mathbf{n}) + \mathbf{R}\mathbf{w}]. \end{aligned} \quad (6b)$$

As the coefficient matrix of this system is positive-definite, it can be solved very efficiently for  $\Delta \mathbf{w}$ .  $\Delta \mathbf{n}$  can then be obtained from (6a).

As according to Lemma 1 the Hessian matrix  $\nabla^2 G$  for problem (1) is only positive, semi-definite range space methods cannot directly be applied to the solution of equations (3). This difficulty can be circumvented, however, if we introduce the total mole numbers  $N^j, j = 1, \dots, p$  of the phases as additional independent variables and define modified free energies:

$$\hat{G}^j(\mathbf{n}') = G^j(\mathbf{n}') + \left( \sum_{i=1}^{s(j)} n'_i \right) \log \left( \sum_{i=1}^{s(j)} n'_i \right).$$

The functions  $\hat{G}^j$  have the following remarkable property, whose proof is given in Proposition A2 in Appendix A.

#### Proposition 2

The Hessian matrices  $\nabla^2 \hat{G}^j(\mathbf{n}'), j = 1, \dots, p$  are positive-definite.

As a matter of fact the addition of any function  $\phi(\sum_{i=1}^{s(j)} n'_i)$  with a positive second derivative turns  $G^j$  into a positive-definite function (see Proposition A2

in Appendix A). Our particular choice has the advantage that the Hessian matrix for an ideal phase is transformed to diagonal form (cf. below and Appendix C).

With the stipulation that:

$$N^j = \sum_{i=1}^{s(j)} n'_i, \quad j = 1, \dots, p, \quad (7)$$

equations (3) can now be written as:

$$\nabla \hat{G}^j(\mathbf{n}') + \mathbf{R}^j \mathbf{w} - \mathbf{E}^j[1 + \log(N^j)] = \mathbf{0}, \quad j = 1, \dots, p, \quad (8a)$$

$$\mathbf{Rn} = \sum_{j=1}^p \mathbf{R}^j \mathbf{n}^j = \mathbf{b}, \quad (8b)$$

$$\hat{\mathbf{E}}^j \mathbf{n}^j - N^j = 0 \quad \text{for } j = 1, \dots, p. \quad (8c)$$

Here  $\mathbf{E}^j$  denotes the  $[s(j), 1]$  matrix with constant entry 1,

$$\mathbf{E}^j = (1, \dots, 1)^T \in R^{s(j)}.$$

Linearization of equations (8) leads to the equations:

$$\begin{aligned} \nabla^2 \hat{G}^j(\mathbf{n}') \Delta \mathbf{n}^j + \mathbf{R}^j \Delta \mathbf{w} - \mathbf{E}^j \frac{\Delta N^j}{N^j} = -\nabla \hat{G}^j(\mathbf{n}') - \mathbf{R}^j \mathbf{w} \\ + \mathbf{E}^j[1 + \log(N^j)] \quad j = 1, \dots, p, \end{aligned} \quad (9a)$$

$$\mathbf{R} \Delta \mathbf{n} = -\mathbf{Rn} + \mathbf{b}, \quad (9b)$$

$$\hat{\mathbf{E}}^j \Delta \mathbf{n}^j - \Delta N^j = -\hat{\mathbf{E}}^j \mathbf{n}^j + N^j, \quad j = 1, \dots, p. \quad (9c)$$

Equations (9c) show that requirement (7) for the initial iterate is passed on to the subsequent iterates. Thus granting (7) the iterates for equations (9) and (3) are identical.

As the matrices  $\nabla^2 \hat{G}^j$  are positive-definite, equations (9a) can be solved for the  $\Delta \mathbf{n}^j$ :

$$\Delta \mathbf{n}^j = -\mathbf{H}^j \left( \mathbf{R}^j \Delta \mathbf{w} - \mathbf{E}^j \frac{\Delta N^j}{N^j} + \mathbf{c}^j \right), \quad j = 1, \dots, p. \quad (10)$$

Here the abbreviations:

$$\mathbf{c}^j = \nabla \hat{G}^j(\mathbf{n}') + \mathbf{R}^j \mathbf{w} - \mathbf{E}^j[1 + \log(N^j)], \quad j = 1, \dots, p,$$

$$\mathbf{c} = (\mathbf{c}^1, \dots, \mathbf{c}^p),$$

$$\mathbf{H}^j = \nabla^2 \hat{G}^j(\mathbf{n}')^{-1} \quad j = 1, \dots, p,$$

$\mathbf{H}$  = diagonal matrix with blocks  $\mathbf{H}^1, \dots, \mathbf{H}^p$ , are used. Introducing expression (10) into equations (9b) and (9c) leads to a system of  $(m+p)$  linear equations in the unknowns  $\Delta \mathbf{w}$  and  $\Delta N^j$ :

$$\begin{aligned} (\mathbf{RH}\mathbf{R})\Delta \mathbf{w} - \sum_{j=1}^p \mathbf{R}^j \mathbf{H}^j \mathbf{E}^j \frac{\Delta N^j}{N^j} \\ = -\mathbf{b} + \mathbf{Rn} - \sum_{j=1}^p \mathbf{R}^j \mathbf{H}^j \mathbf{c}^j, \end{aligned} \quad (11a)$$

$$\begin{aligned} (\hat{\mathbf{E}}^j \mathbf{H}^j \mathbf{R}^j) \Delta \mathbf{w} - (\hat{\mathbf{E}}^j \mathbf{H}^j \mathbf{E}^j - N^j) \frac{\Delta N^j}{N^j} \\ = -N^j + \hat{\mathbf{E}}^j \mathbf{n}^j - \hat{\mathbf{E}}^j \mathbf{H}^j \mathbf{c}^j, \quad j = 1, \dots, p. \end{aligned} \quad (11b)$$

Equations (11) can now be solved for the  $\Delta w$  and  $\Delta N^j, j = 1, \dots, p$ . The corrections to the mole numbers  $\Delta n^j, j = 1, \dots, p$  are then obtained from equations (10).

The ideal approximation method proceeds in the same way but neglects the excess contribution  $G_{ex}^j$  to the functions  $\hat{G}^j$ . Thus:

$$\hat{G}^j(n^j) = \sum_{i=1}^{s(j)} n_i^j \log(n_i^j)$$

and therefore the  $\nabla^2 \hat{G}^j, j = 1, \dots, p$  are diagonal matrices with entries  $1/n_i^j, i = 1, \dots, s(j)$  which can readily be inverted.

Defining the  $(m, p)$  matrix:

$$\mathbf{B} = (\mathbf{R}^1 \mathbf{H}^1 \mathbf{E}^1, \dots, \mathbf{R}^p \mathbf{H}^p \mathbf{E}^p),$$

the  $(p, p)$ -matrix

$\mathbf{F}$  = diagonal matrix with entries

$$F_{jj} = \tilde{\mathbf{E}}^j \tilde{\mathbf{H}}^j \tilde{\mathbf{E}}^j - N^j = \sum_{i=1}^{s(j)} \mathbf{H}_{k,i}^j - N^j, \quad j = 1, \dots, p,$$

the vectors

$$\mathbf{N} = (N^1, \dots, N^p) \quad \text{and}$$

$$\Delta \log(\mathbf{N}) = \left( \frac{\Delta N^1}{N^1}, \dots, \frac{\Delta N^p}{N^p} \right)$$

and

$$\mathbf{g} = (-N^1 + \tilde{\mathbf{E}}^1 \mathbf{n}^1 - \tilde{\mathbf{E}}^1 \mathbf{H}^1 \mathbf{c}^1, \dots, -N^p + \tilde{\mathbf{E}}^p \mathbf{n}^p - \tilde{\mathbf{E}}^p \mathbf{H}^p \mathbf{c}^p),$$

equations (11) can be written in a more compact form as

$$\begin{pmatrix} \mathbf{R} \mathbf{H} \tilde{\mathbf{R}} - \mathbf{B} \\ \mathbf{B} - \mathbf{F} \end{pmatrix} \begin{pmatrix} \Delta \mathbf{w} \\ \Delta \log(\mathbf{N}) \end{pmatrix} = \begin{pmatrix} -\mathbf{b} + \mathbf{R} \mathbf{n} - \mathbf{R} \mathbf{H} \mathbf{c} \\ \mathbf{g} \end{pmatrix}. \quad (12)$$

The elimination procedure does not necessarily have to be applied to all solution phases. If for example only the increments of the first  $t$  phases  $\Delta n^j, j = 1, \dots, t$  are eliminated from equations (9) by means of formula (10) the following system of equations for the unknowns  $\Delta \mathbf{w}, \Delta N^j, j = 1, \dots, t$  and  $\Delta n^{t+1}, \dots, \Delta n^p$  is obtained:

$$\sum_{j=1}^t (\mathbf{R}^j \mathbf{H}^j \tilde{\mathbf{R}}^j) \Delta \mathbf{w} - \sum_{j=1}^t \mathbf{R}^j \mathbf{H}^j \mathbf{E}^j \frac{\Delta N^j}{N^j} - \sum_{j=t+1}^p \mathbf{R}^j \Delta n^j = -\mathbf{b} + \mathbf{R} \mathbf{n} + \sum_{j=1}^t \mathbf{R}^j \mathbf{H}^j \mathbf{c}^j, \quad (13a)$$

$$(\tilde{\mathbf{E}}^j \mathbf{H}^j \tilde{\mathbf{R}}^j) \Delta \mathbf{w} - (\tilde{\mathbf{E}}^j \tilde{\mathbf{H}}^j \mathbf{E}^j - N^j) \frac{\Delta N^j}{N^j} = -N^j + \tilde{\mathbf{E}}^j \mathbf{n}^j - \tilde{\mathbf{E}}^j \mathbf{H}^j \mathbf{c}^j, \quad (13b)$$

for  $j = 1, \dots, t$  and

$$\tilde{\mathbf{R}}^j \Delta \mathbf{w} + \nabla^2 G^j(n^j) \Delta n^j = -\nabla G^j(n^j) - \tilde{\mathbf{R}}^j \mathbf{w}, \quad (13c)$$

for  $j = t+1, \dots, p$ .

Once the variables  $\Delta \mathbf{w}, \Delta N^j, j = 1, \dots, t$  and  $\Delta n^j, j = t+1, \dots, p$  have been calculated from equations (13), the remaining  $\Delta n^j, j = 1, \dots, t$  are determined by equations (10).

As an example for partial elimination consider a system consisting of a gas phase with many species and several condensed solution phases. The free energy  $G$  of the gas phase has a dense matrix of second partial derivatives, whereas the matrix of second partial derivatives of the modified free energy  $\hat{G}$  is diagonal (cf. Appendix C). In this case the elimination of the gas phase variables already leads to a considerable reduction in computing cost for the solution of equations (4).

In Appendix C we present the operation counts for the various methods for solving the Newton equations (4). To determine the most economical method for a given problem one simply has to evaluate and compare the operation counts. For the null space method the cost of providing the matrix  $\mathbf{Z}$ , which has to be formed only once, is disregarded. One should also bear in mind the storage requirements of the direct and null space method.

The analysis shows that for  $n$  much larger than  $m$  the range space method is by far the most economical one, in particular for systems with a large ideal gas phase. For systems with  $n$  about equal to  $m$  the null space method is to be preferred. In general the cost of a solution method depends on  $n, m, p$  and  $s(j), j = 1, \dots, p$  and more details are given in Appendix C. For systems with only a small number of component species  $n$  the difference in cost between the direct, null space and range space method is only marginal and for reasons of numerical stability the direct solution method is recommended (cf. Heath, 1978).

#### 4. NUMERICAL IMPLEMENTATION

To calculate the Newton correction  $\Delta \mathbf{w}$  and  $\Delta N^j/N^j, j = 1, \dots, p$  one first has to invert the positive-definite matrices  $\nabla^2 \hat{G}^j$  for each phase  $j = 1, \dots, p$ . This can be readily accomplished by factoring each block by Cholesky's method. One then has to compute the coefficients of the linear equations for  $\Delta \mathbf{w}$  and  $-\Delta N^j/N^j$ :

$$(\mathbf{R} \mathbf{H} \tilde{\mathbf{R}}) \Delta \mathbf{w} + \sum_{j=1}^p \mathbf{R}^j \mathbf{H}^j \mathbf{E}^j \left( -\frac{\Delta N^j}{N^j} \right) = -\mathbf{b} + \mathbf{R} \mathbf{n} - \sum_{j=1}^p \mathbf{R}^j \mathbf{H}^j \mathbf{c}^j, \quad (14a)$$

$$(\tilde{\mathbf{E}}^j \mathbf{H}^j \tilde{\mathbf{R}}^j) \Delta \mathbf{w} + (\tilde{\mathbf{E}}^j \tilde{\mathbf{H}}^j \mathbf{E}^j - N^j) \left( -\frac{\Delta N^j}{N^j} \right) = -N^j + \tilde{\mathbf{E}}^j \mathbf{n}^j - \tilde{\mathbf{E}}^j \mathbf{H}^j \mathbf{c}^j, \quad j = 1, \dots, p. \quad (14b)$$

As the linear system (14) has a symmetric coefficient matrix, it can be solved by a special routine which compared to Gaussian elimination halves the computational cost (cf. Dongarra, 1979). Numerical difficulties arise if the coefficient matrix in equations (14) is nearly singular. According to Proposition 1 this occurs if the vectors  $\mathbf{R}^j \mathbf{n}^j, j = 1, \dots, p$  become

nearly linearly dependent. In this case one should try to determine phases  $j_1, \dots, j_r$  such that the vectors:

$$\mathbf{R}^{j_1} \mathbf{n}^{j_1}, \dots, \mathbf{R}^{j_r} \mathbf{n}^{j_r},$$

form a well-conditioned basis of the linear space determined by the  $\mathbf{R}^j \mathbf{n}^j$ ,  $j = 1, \dots, p$ . The equilibrium of these phases then furnishes Lagrange multipliers  $\mathbf{w}$ , from which the compositions of the remaining phases can be calculated from the equations:

$$\nabla G^j(\mathbf{n}^j) + \mathbf{R}^j \mathbf{w} = 0.$$

Furthermore it is advisable to use a representation of equations (1b) in which the column vectors in the constraint matrix  $\mathbf{R}$  corresponding to the most abundant species become unit vectors (Smith and Missen, 1982; Schnedler, 1984).

Computational experience has also demonstrated that for nearly degenerate minima the generalized linear programming algorithm described in Greiner (1988b) can furnish a very accurate estimate of the solution, which can either be considered as sufficiently precise or further improved by Newton's method.

The matrices  $\nabla^2 G^j(\mathbf{n})$  can be calculated from the Hessian matrices of the molar free energies  $\nabla^2 \bar{G}^j$  with the help of the formula given in Lemma A1 in Appendix A. The second derivatives of the molar free energies can be obtained either analytically or from first- or second-order finite differences from the analytical derivatives or the function values, respectively.

## 5. CONCLUSIONS

We have presented an efficient and reliable numerical implementation of the Brinkley-Nasa-Rand algorithm for the calculation of general multiphase chemical equilibria involving nonideal phases. It overcomes the convergence problems associated with the ideal approximation method and can handle nonideal systems of any complexity effectively. As any Newton method, our algorithm requires a good initial estimate of the solution as a starting point. This estimate can be obtained by an algorithm based on generalized linear programming.

Our computational experience has so far vindicated our theoretical claim that this approach can deal with the most complex systems efficiently and reliably [cf. Appendix B and Greiner and Schnedler (1991)].

**Acknowledgements**—The author would like to thank Bernd Rüber and the referees for helpful suggestions.

## NOMENCLATURE

- $b_i$  = Mole number of the  $i$ th chemical element present in the system ( $i = 1, \dots, m$ )  
 $\mathbf{b} = (b_1, \dots, b_m)$   
 $\mathbf{c}^j = \nabla \bar{G}^j(\mathbf{n}^j) + \mathbf{R}^j \mathbf{w} - \mathbf{E}^j [1 + \log(N^j)]$ ,  $j = 1, \dots, p$   
 $\mathbf{c} = (\mathbf{c}^1, \dots, \mathbf{c}^p)$   
 $\mathbf{D}^j$  = Diagonal  $[s(j), s(j)]$  matrix with entries  $n_i^j$ ,  $j = 1, \dots, p$

- $\mathbf{E}^j = [s(j), 1]$  matrix with constant entry 1,  $j = 1, \dots, p$   
 $G_{ex}^j$  = Free excess energy of  $j$ th phase,  $j = 1, \dots, p$   
 $\bar{G}^j$  = Molar free energy of  $j$ th phase,  $j = 1, \dots, p$   
 $G^j$  = Free energy of  $j$ th phase,  $j = 1, \dots, p$   
 $G$  = Total free energy  
 $\bar{G}^j = \bar{G}^j(\mathbf{n}^j) = \bar{G}^j(\mathbf{n}^j) + (\sum_{i=1}^{s(j)} n_i^j) \log(\sum_{i=1}^{s(j)} n_i^j)$ ,  $j = 1, \dots, p$   
 $\mathbf{H}^j$  = Matrix inverse of  $\nabla^2 \bar{G}^j$ ,  $j = 1, \dots, p$   
 $\mathbf{H}$  = Block diagonal matrix with blocks  $\mathbf{H}^j$ ,  $j = 1, \dots, p$   
 $i$  = Subscript numbering component species  
 $\mathbf{I}$  = Identity matrix  
 $j$  = Superscript numbering solution phases  
 $m$  = Number of chemical elements  
 $n$  =  $\sum_{j=1}^p s(j)$  total number of species  
 $N^j$  = Total mole number of  $j$ th phase,  $j = 1, \dots, p$   
 $n_i^j$  = Mole number of  $i$ th species in  $j$ th phase  
 $\mathbf{n}^j = (n_1^j, \dots, n_{s(j)}^j)$  vector of species mole numbers in  $j$ th phase  
 $p$  = Number of solution phases  
 $\mathbf{q} = \text{Log}(\mathbf{n})$   
 $\mathbf{R}_i^j$  = Column vector with  $m$  components. The  $k$ th component gives the number of moles of element  $k$  in one mole of species  $i$  in phase  $j$   
 $\mathbf{R}^j = (\mathbf{R}_1^j, \dots, \mathbf{R}_{s(j)}^j)$ ,  $[m, s(j)]$   
 $\mathbf{R} = (\mathbf{R}^1, \dots, \mathbf{R}^p)$ ,  $[m, n]$  matrix  
 $s(j)$  = Number of component species in  $j$ th phase,  $j = 1, \dots, p$   
 $\sim$  = Matrix transposition  
 $V_0^j = \{v \in \mathbf{R}^j | \sum_{i=1}^{s(j)} v_i = 0\}$   
 $\mathbf{w} = (w_1, \dots, w_m)$  Lagrange multipliers  
 $\mathbf{x} = (x_1, \dots, x_s)$  mole fractions  
 $\mathbf{X} = (s, s)$  matrix with identical rows  $(x_1, \dots, x_s)$   
 $\mathbf{Z} = (n, n - m)$  matrix describing the null space of the matrix  $\mathbf{R}$

## Greek symbols

- $\delta_{ij}$  = Kronecker's symbol  
 $\nabla f$  = gradient of function  $f$   
 $\nabla^2 f$  = Matrix of second derivatives of  $f$   
 $\mu^0$  = Chemical potential of a pure substance

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## APPENDIX A

### Mathematical Proofs

#### Lemma A1

The Hessian matrices of the Gibbs free energy  $G$  and the molar free energy  $\bar{G}$  of a phase are related by:

$$N\nabla_n^2 G(\mathbf{n}) = (\mathbf{I} - \mathbf{X})\nabla_x^2 \bar{G}(\mathbf{x})(\mathbf{I} - \mathbf{X}),$$

where  $\mathbf{X}$  denotes the  $(s, s)$  matrix with identical row vectors  $(x_1, \dots, x_s)$  and  $N = \sum_{i=1}^s n_i$ .

**Proof.** Differentiating:

$$x_i = \frac{n_i}{\sum_{j=1}^s n_j},$$

gives the relations

$$\frac{\partial x_j}{\partial n_i} = \frac{1}{N} (\delta_{ij} - x_j), \quad (\text{A1})$$

which will be used repeatedly.

From the definition of  $G$  we have:

$$\begin{aligned} \frac{\partial G}{\partial n_k} &= \bar{G}(x_1, \dots, x_s) + N \sum_{i=1}^s \frac{\partial \bar{G}}{\partial x_i} \frac{\partial x_i}{\partial n_k} \\ &= \bar{G}(x_1, \dots, x_s) - \sum_{i=1}^s \frac{\partial \bar{G}}{\partial x_i} x_i + \frac{\partial \bar{G}}{\partial x_k}. \end{aligned}$$

Hence:

$$\begin{aligned} \frac{\partial^2 G}{\partial n_k \partial n_l} &= \sum_{i=1}^s \frac{\partial \bar{G}}{\partial x_i} \frac{\partial^2 x_i}{\partial n_k \partial n_l} - \sum_{ij=1}^s x_i \frac{\partial^2 \bar{G}}{\partial x_i \partial x_j} \frac{\partial x_j}{\partial n_l} \\ &\quad - \sum_{i=1}^s \frac{\partial \bar{G}}{\partial x_i} \frac{\partial^2 x_i}{\partial x_k \partial n_l} + \sum_{i=1}^s \frac{\partial^2 \bar{G}}{\partial x_k \partial x_i} \frac{\partial x_i}{\partial n_l}. \end{aligned}$$

Here the first and third term on the right-hand side cancel. Introduction of the expression (A1) into the remaining terms and rearrangement of the sums then leads to:

$$\begin{aligned} N \frac{\partial^2 G}{\partial n_k \partial n_l} &= \frac{\partial^2 \bar{G}}{\partial x_k \partial x_l} + \sum_{ij=1}^s x_i \frac{\partial^2 \bar{G}}{\partial x_i \partial x_j} x_j \\ &\quad - \sum_{i=1}^s x_i \frac{\partial^2 \bar{G}}{\partial x_i \partial x_l} - \sum_{i=1}^s x_i \frac{\partial^2 \bar{G}}{\partial x_i \partial x_k}, \end{aligned}$$

which expressed in matrix notation gives the result.

#### Lemma A2

The matrix operator  $\mathbf{I} - \mathbf{X}: R' \rightarrow R'$  has the following easily verifiable properties: let  $\mathbf{v} = (\mathbf{I} - \mathbf{X})\mathbf{w}$  with vectors  $\mathbf{v}$ ,  $\mathbf{w}$  from  $R'$ . Then:

- (1)  $\mathbf{v} \in V_0'$ , i.e.  $\sum_{i=1}^s v_i = 0$ ;
- (2)  $\mathbf{v} = 0$  if and only if  $\mathbf{w} = \lambda \mathbf{n}$  for some scalar  $\lambda$ .

**Proof.** This follows immediately from:

$$v_i = w_i - x_i \left( \sum_{k=1}^s w_k \right), \quad i = 1, \dots, s.$$

As a consequence of Lemmas A1 and A2 we obtain:

#### Lemma A3

The free energy  $G$  of a phase is positive semi-definite and  $\bar{\mathbf{w}}\nabla^2 G(\mathbf{n})\mathbf{w} = 0$  if and only if  $\mathbf{w} = \lambda \mathbf{n}$  for some scalar  $\lambda$ .

**Proof.** This follows from the representation:

$$\bar{\mathbf{w}}N\nabla_n^2 G(\mathbf{n})\mathbf{w} = \bar{\mathbf{w}}(\mathbf{I} - \mathbf{X})\nabla_x^2 \bar{G}(\mathbf{x})(\mathbf{I} - \mathbf{X})\mathbf{w}$$

and the assumptions on  $\bar{G}$ .

#### Proposition A1

**Proof of Proposition 1:** to establish the equivalence of statements 1 and 3 consider the minimization problem:

$$\frac{1}{2}(\Delta \mathbf{n})\nabla^2 G(\mathbf{n})\Delta \mathbf{n} = \min \text{ subject to } \mathbf{R}\Delta \mathbf{n} = 0,$$

where  $G$  is the total free energy. As  $\nabla^2 G$  is positive semi-definite this problem is well-defined and the necessary and sufficient condition for a minimum  $\Delta \mathbf{n}$  is the existence of multipliers  $\mathbf{w}$  such that:

$$[\nabla^2 G(\mathbf{n})]\Delta \mathbf{n} + \bar{\mathbf{R}}\mathbf{w} = 0 \quad \text{and} \quad \mathbf{R}\Delta \mathbf{n} = 0.$$

Hence these equations have the unique solution  $\Delta \mathbf{n} = 0$  and  $\mathbf{w} = 0$  if and only if  $\nabla^2 G$  is positive-definite on the subspace:

$$\mathbf{W} = \{\Delta \mathbf{n} | \mathbf{R}\Delta \mathbf{n} = 0\}.$$

To show that statements 2 and 3 are equivalent assume that there exists a  $\Delta \mathbf{n} \neq 0$  satisfying:

$$\Delta \mathbf{n}\nabla^2 G(\mathbf{n})\Delta \mathbf{n} = \sum_{j=1}^p \Delta \mathbf{n}^j \nabla^2 G^j(\mathbf{n}^j)\Delta \mathbf{n}^j = 0$$

and

$$\mathbf{R}\Delta \mathbf{n} = \sum_{j=1}^p \mathbf{R}^j \Delta \mathbf{n}^j = 0.$$

As the  $\nabla^2 G^j$  are positive semi-definite we have:

$$\Delta \mathbf{n}^j \nabla^2 G^j(\mathbf{n}^j)\Delta \mathbf{n}^j = 0, \quad j = 1, \dots, p$$

and hence  $(\mathbf{I} - \bar{\mathbf{X}}^{(j)})\Delta \mathbf{n}^j = 0$ . This entails  $\Delta \mathbf{n}^j = \lambda_j \mathbf{n}^j$  for  $j = 1, \dots, p$  and therefore the vectors  $\mathbf{R}^j \mathbf{n}^j$  are linearly dependent:

$$\mathbf{R}\Delta \mathbf{n} = \sum_{j=1}^p \lambda_j \mathbf{R}^j \mathbf{n}^j = 0.$$

( $\Delta \mathbf{n} \neq 0$  requires that  $\lambda_j \neq 0$  for at least one  $j$ !). Reversing the argument gives the converse implication.

#### Proposition A2

Assume that  $\bar{G}(x_1, \dots, x_s)$  is strictly convex. Define:

$$\begin{aligned} \hat{G}(n_1, \dots, n_s) &= G(n_1, \dots, n_s) + F(n_1, \dots, n_s) \\ &= \left( \sum_{i=1}^s n_i \right) \bar{G}(x_1(\mathbf{n}), \dots, x_s(\mathbf{n})) + \phi \left( \sum_{i=1}^s n_i \right), \end{aligned}$$

where  $\phi$  is a real function with a strictly positive second derivative. Then  $\nabla^2 \hat{G}$  is positive-definite.

**Proof.**  $\nabla^2 F$  is a matrix with constant entry:

$$a = \phi'' \left( \sum_{i=1}^s n_i \right),$$

and therefore positive-definite. The matrix  $\nabla^2 \hat{G} = \nabla^2 G + \nabla^2 F$  being the sum of two positive semi-definite matrices is clearly positive semi-definite. To prove that it is in fact positive-definite assume that  $\bar{\mathbf{w}}\nabla^2 \hat{G}\mathbf{w} = 0$  with  $\mathbf{w} \neq 0$  and hence  $\mathbf{w} = \lambda \mathbf{n}$  with  $\lambda \neq 0$ . It follows:

$$\bar{\mathbf{w}}\nabla^2 F\mathbf{w} = \lambda^2 a \sum_{i,k=1}^s n_k n_i = \lambda^2 a \left( \sum_{i=1}^s n_i \right)^2 > 0.$$

## APPENDIX B

*Convergence Problems of the Ideal Approximation Method*

The following examples are meant to exhibit the convergence problems of the ideal approximation method. For illustrative purposes we have chosen a particularly simple system involving two binary solution phases. It can be calculated by the null-space method and clearly demonstrates the deficiencies of the ideal approximation method, which occur for the null-space and the range-space method alike. For more complex systems for which we have experienced similar convergence problems see Greiner and Schnedler (1991).

The equilibrium mole numbers have been obtained by first determining a close starting point for Newton's method by the generalized linear programming method described in Greiner (1988b). We then applied the range-space method as developed in this paper. When not explicitly stated we did not perform line search [see e.g. Dennis and Schnabel (1983) Chap. 6].

We consider a system consisting of two compounds A and B which exist in the gaseous and liquid state. We stipulate that the ideal gas phase A(g) and B(g) exists at a pressure of 1 bar and that A(l) and B(l) form a regular solution with an interaction parameter (in units of RT) of  $-3$ . The chemical potentials of the species (in units of RT) are assumed at follows:

$$\mu_{A(g)} = -53.56926553, \quad \mu_{B(g)} = -54.33693060,$$

$$\mu_{A(l)} = -54.25814819, \quad \mu_{B(l)} = -51.50413543.$$

The system contains 0.37249 mol of compound A and 0.12749 mol of compound B.

The exact equilibrium solution for this system is given by:

$$n_{A(g)} = 0.0085794521, \quad n_{B(g)} = 0.0175663182,$$

$$n_{A(l)} = 0.3639205492, \quad n_{B(l)} = 0.1099336847.$$

When the exact Newton algorithm was started from the point:

$$n_{A(g)} = 0.0082908617, \quad n_{B(g)} = 0.0169754444,$$

$$n_{A(l)} = 0.3647111606, \quad n_{B(l)} = 0.1100225333.$$

convergence was achieved in three iterations. With the same starting point the ideal approximation method did not converge at all, but ran initially away from the solution and then exhibited oscillatory behaviour. When the ideal approximation method was used in conjunction with a line search procedure convergence was achieved, but was extremely slow (about 30 iterations).

In another test we changed the regular interaction parameter to  $-2$  and  $\mu_{B(l)}$  to  $-52.15704744$ . For this system the exact equilibrium mole numbers are:

$$n_{A(g)} = 0.0069858134, \quad n_{B(g)} = 0.0134531341,$$

$$n_{A(l)} = 0.3655141871, \quad n_{B(l)} = 0.1140468669.$$

With a starting point:

$$n_{A(g)} = 0.009000785, \quad n_{B(g)} = 0.017334993,$$

$$n_{A(l)} = 0.360089220, \quad n_{B(l)} = 0.113574990,$$

the exact Newton method needed three iterations to converge, whereas the ideal approximation method took about 45 steps (with and without line search). When the value of  $\mu_{B(l)}$  was changed to  $-49.95776488$  both methods took about the same number of steps from a close starting point.

These realistic examples clearly indicate the shortcomings of the ideal approximation method: even from very close estimates of the true solution it may not converge at all. In other cases its convergence is extremely slow compared to the exact Newton method.

## APPENDIX C

*The Most Efficient Method for Newton's Equations*

In this appendix we give the operation counts for the various methods for solving Newton's equations (4). As usual one operation amounts to one multiplication/division and one addition/subtraction. In counting only the leading third order terms in  $n$ ,  $m$ ,  $p$  and  $s(j)$ ,  $j = 1, \dots, p$  are considered. For more details see Heath (1978) and Golub and van Loan (1984).

*Direct method*

As system (4) is symmetric, its direct solution necessitates  $(n + m)^3/6$  operations.

*Null space methods*

The cost of calculating the matrix  $Z$  from the matrix  $R$  amounts to:

$$nm^2 - \frac{m^3}{2},$$

operations (cf. Heath, 1979).

The calculation of the matrix  $ZV^2GZ$  requires about:

$$n \frac{(n-m)^2}{2} + \sum_{j=1}^p s(j)^2(n-m),$$

operations (here the block diagonal structure of the matrix  $V^2G$  and the symmetry of the matrix  $ZV^2GZ$  have been taken into account).

Finally the factorization of the symmetric matrix  $ZV^2GZ$  takes about:

$$\frac{(n-m)^3}{6},$$

operations.

*Range space method*

In evaluating the cost of the range space method the following facts concerning ideal phases have to be kept in mind: for an ideal phase with Gibbs energy:

$$G(n_1, \dots, n_s) = \sum_{i=1}^s \mu_i^0 n_i + n_i \log \left( \frac{n_i}{N} \right),$$

the matrix

$$(V^2G)_{kl} = \frac{\delta_{kl}}{n_k} - \frac{1}{N},$$

is dense, whereas the modified matrix  $V^2\hat{G}$  is diagonal. Clearly this has to be reckoned with. To simplify the formulae for the operation counts given below, we define:

$$\bar{s}(j) = s(j) \text{ if phase } j \text{ is nonideal,}$$

$$\bar{s}(j) = 0 \text{ if phase } j \text{ is ideal,}$$

and similarly

$$\bar{\bar{s}}(j) = s(j) \text{ if phase } j \text{ is nonideal,}$$

$$\bar{\bar{s}}(j) = \sqrt{s(j)} \text{ if phase } j \text{ is ideal.}$$

The Cholesky factorization of the matrix  $V^2\hat{G}$  needs about:

$$\sum_{j=1}^p \frac{\bar{s}(j)^3}{6},$$

operations.

To form the reduced coefficient matrix (11) one then calculates the matrices  $H/\bar{R}^j$ ,  $j = 1, \dots, p$  which requires about:

$$m \sum_{j=1}^p \bar{s}(j)^2,$$

operations. The symmetric matrix  $RH\bar{R}$  requires another:

$$\frac{n}{2} m^2,$$

operations. The matrices  $\mathbf{E}'\mathbf{H}'\mathbf{R}', j = 1, \dots, p$  and  $\mathbf{E}'\mathbf{H}'\mathbf{E}', j = 1, \dots, p$  cost: Finally the factorization of matrix (11) needs:

$$mp \sum_{j=1}^p s(j) \quad \text{and} \quad \sum_{j=1}^p \bar{s}(j), \quad \frac{(m+p)^3}{6},$$

operations, respectively.

operations.